PREDICTION OF HOMEOWNERSHIP USING LOGISTIC REGRESSION

BY
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Introduction

Objectives

- Analyze U.S. homeownership market trends
- Build a predictive model for homeownership rate

Dataset

- Extract from the **2023 American Community Survey (ACS)**
- Dataset covers homeownership, demographics, geographic and social variables

Background: American Community Survey

Demographic Data:

- Age, Sex, Race, Hispanic origin, Citizenship

Socioeconomic Data:

- Educational attainment, Marital status, School attendance, Total personal income (inctot), Employment status, Class of worker, Hours worked/week

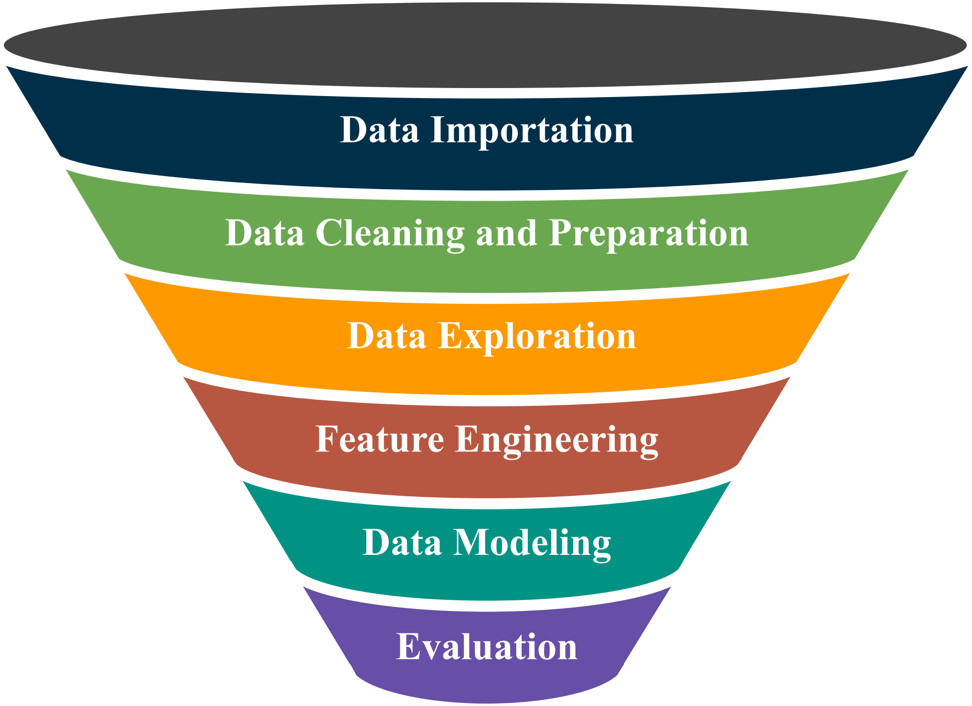
Housing Data:

- Household income, Metropolitan Status, Statefip

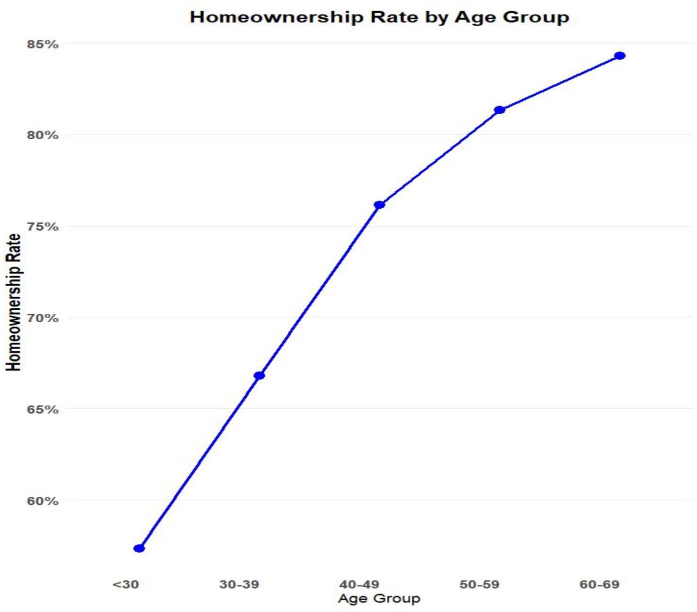
Dependent Variable:

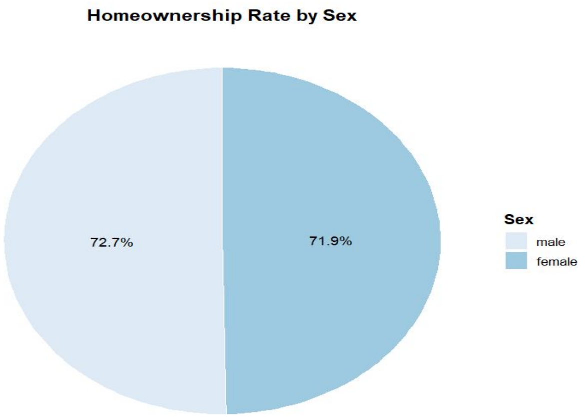
- Homeownership (ownershp): 0 – rented, 1. owned or being bought (loan)

Methodology

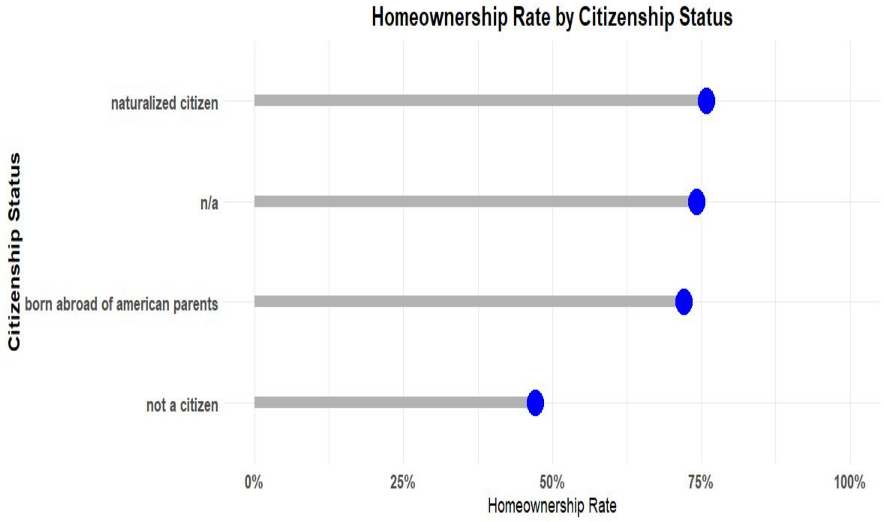


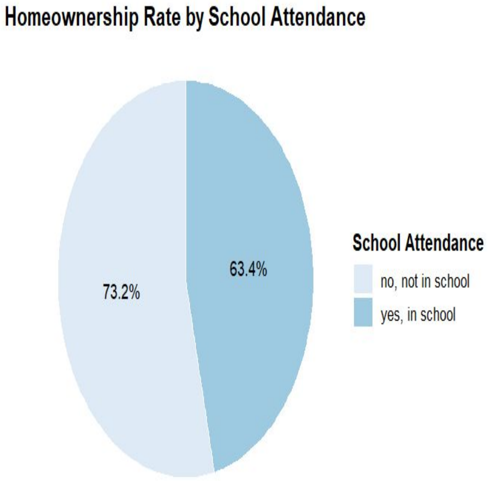
Data Exploration: Demographics



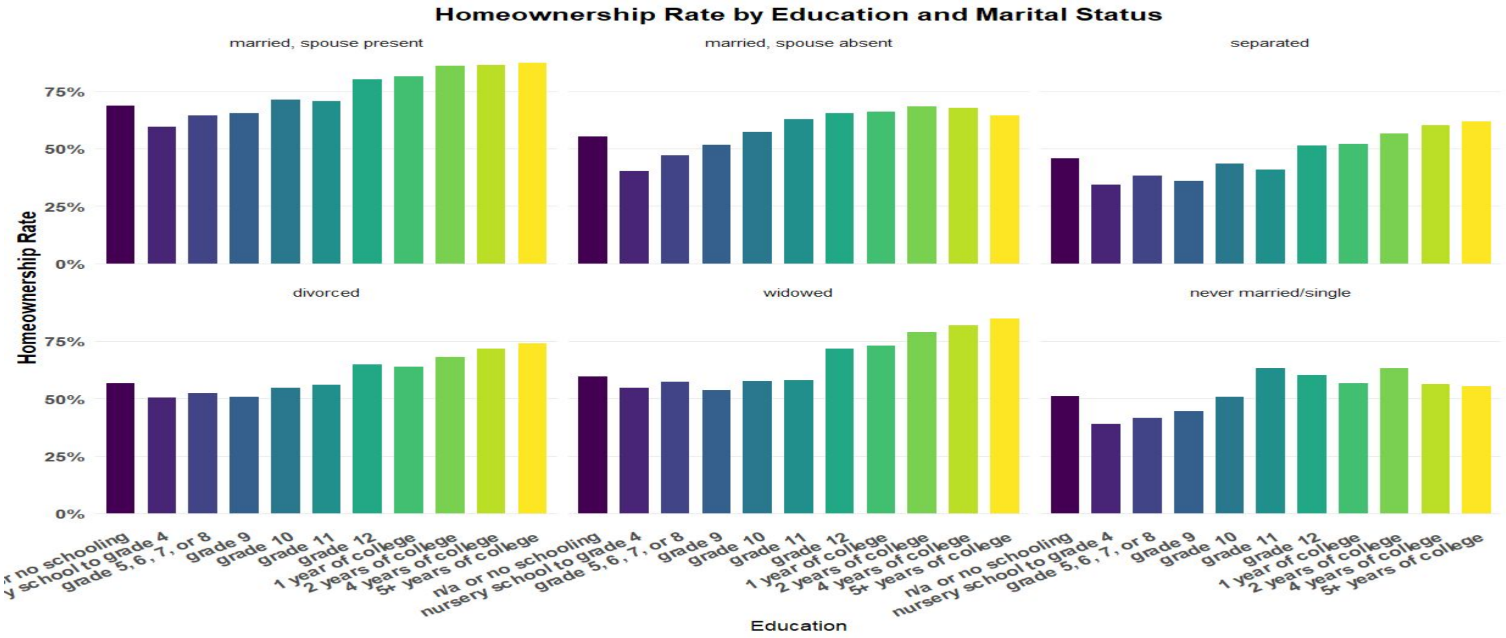


Data Exploration: Socioeconomic

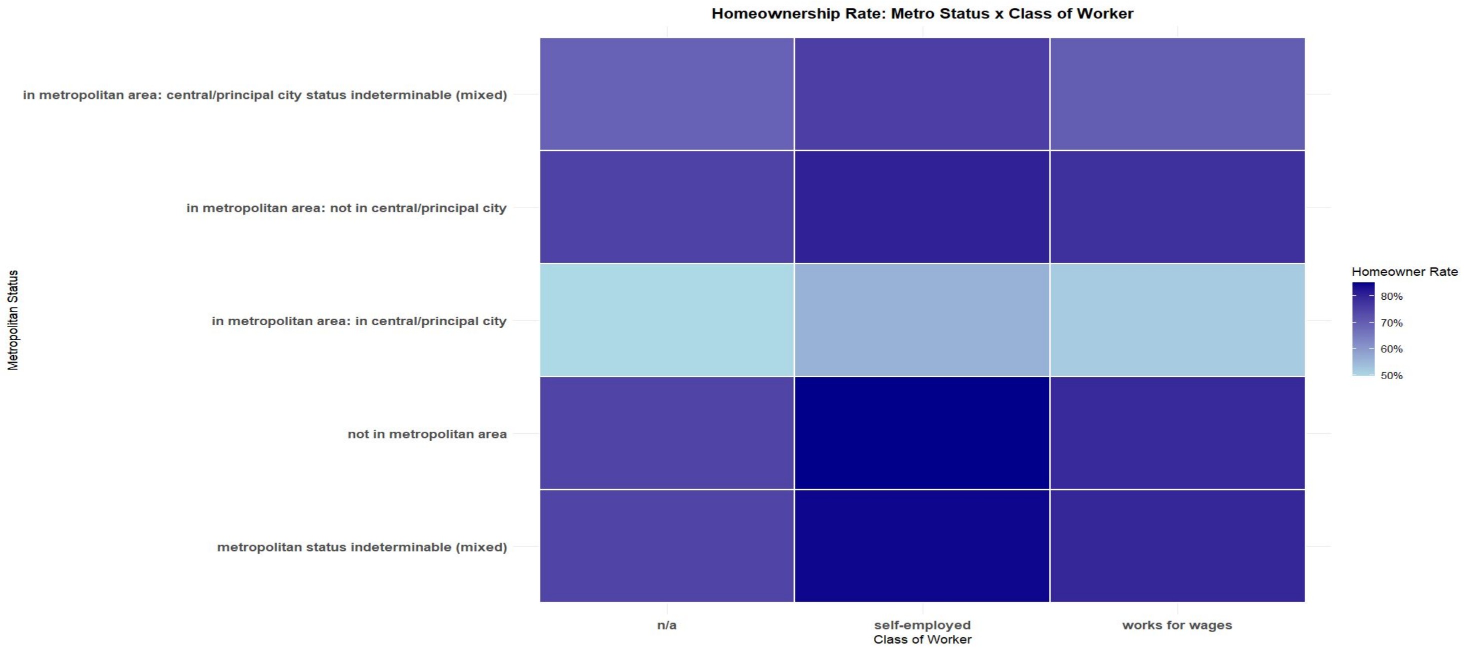




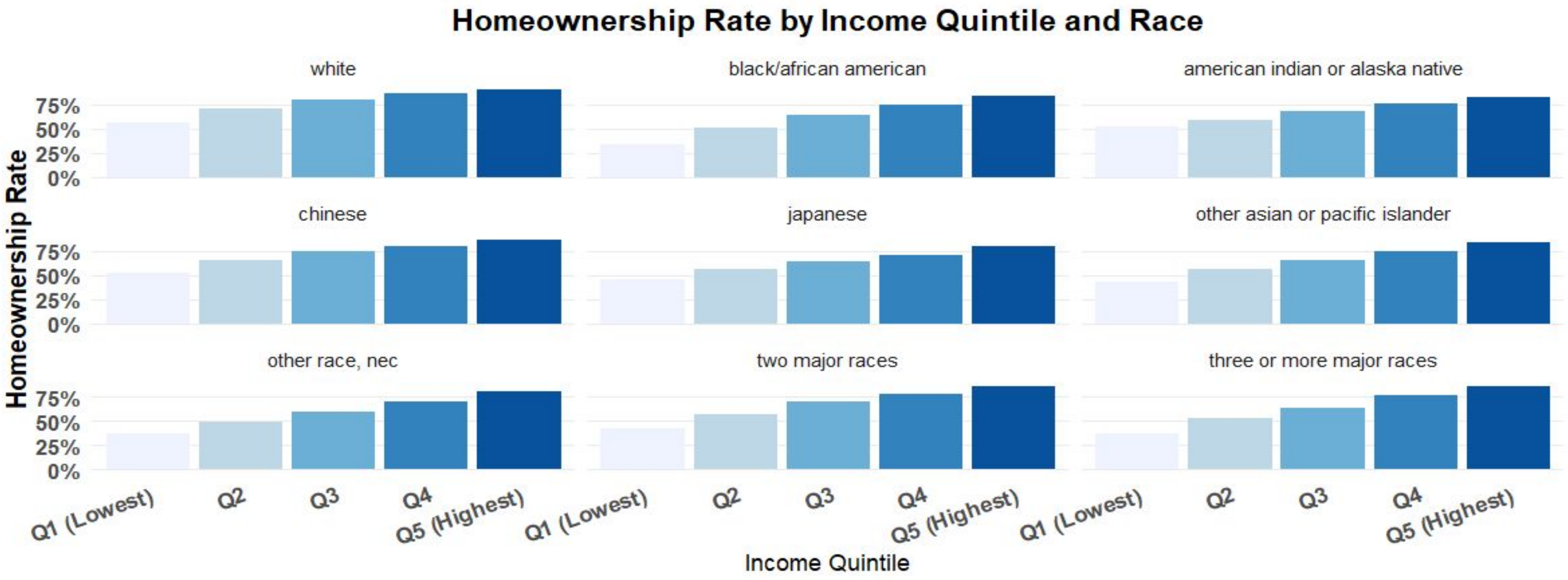
Data Exploration: Socioeconomic



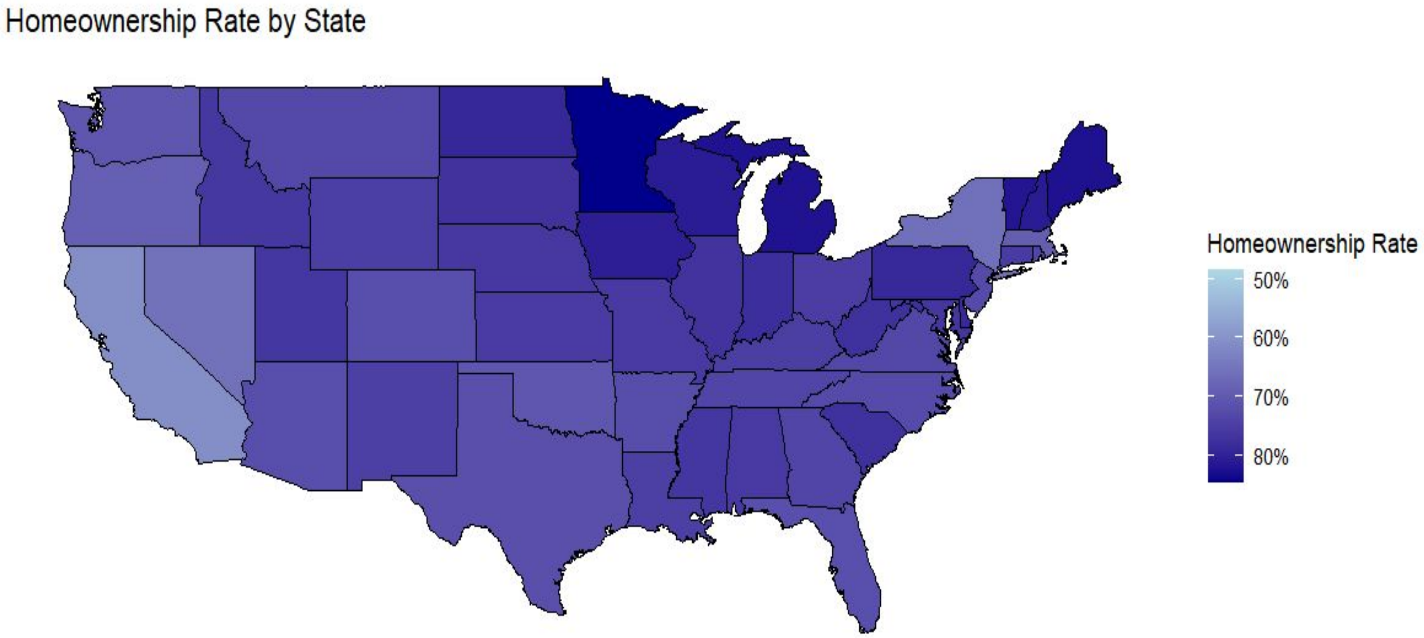
Data Exploration: Housing



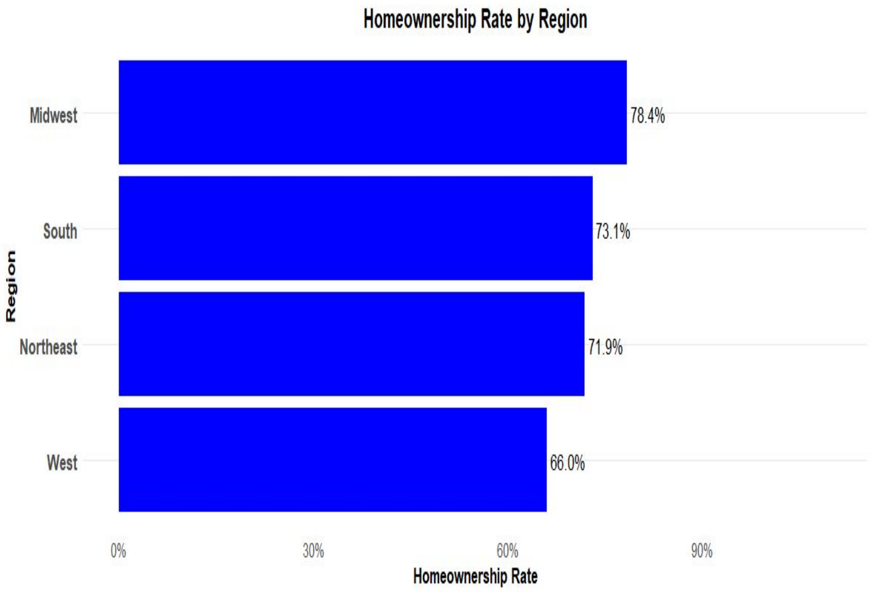
Data Exploration: Housing x Demo

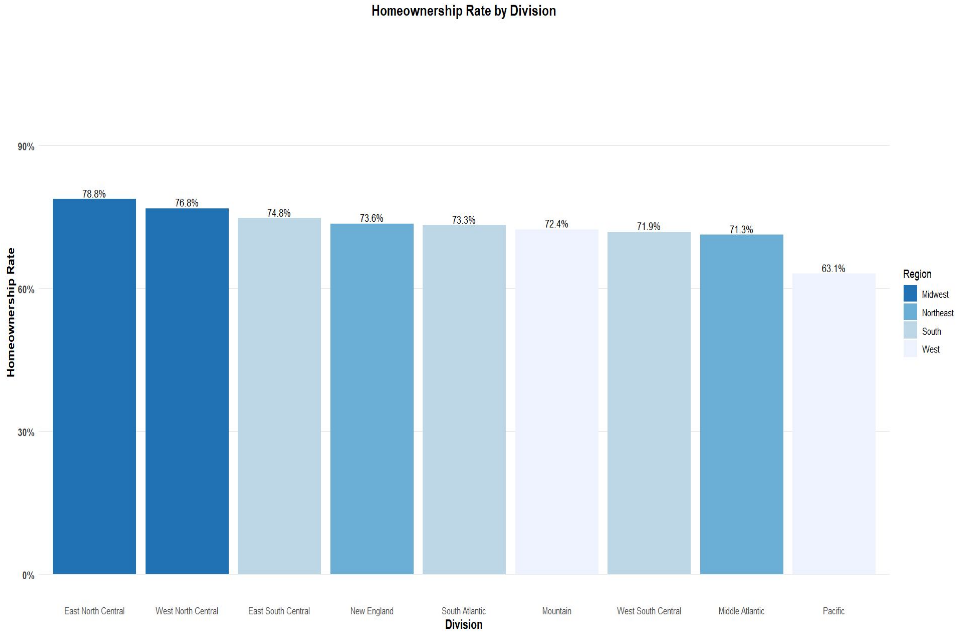


Data Exploration: Geographic

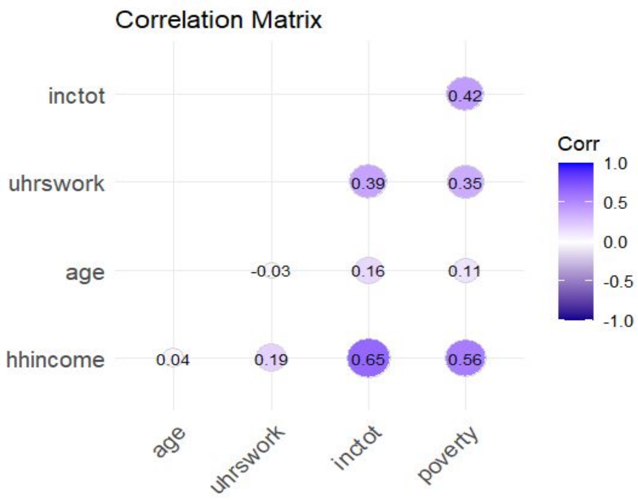


Data Exploration: Geographic





Correlation Analysis



Data Modeling: Logistic Regression

Accuracy:

- Model achieved accuracy of **76.92%** validating its ability to correctly predict outcome

AUC (Area Under the Curve):

- AUC value of **0.7863,** indicates model has good discriminatory power
- Model has **78.63% confidence** in classifying homeowners from non-homeowners

Data Modeling: Logistic Regression

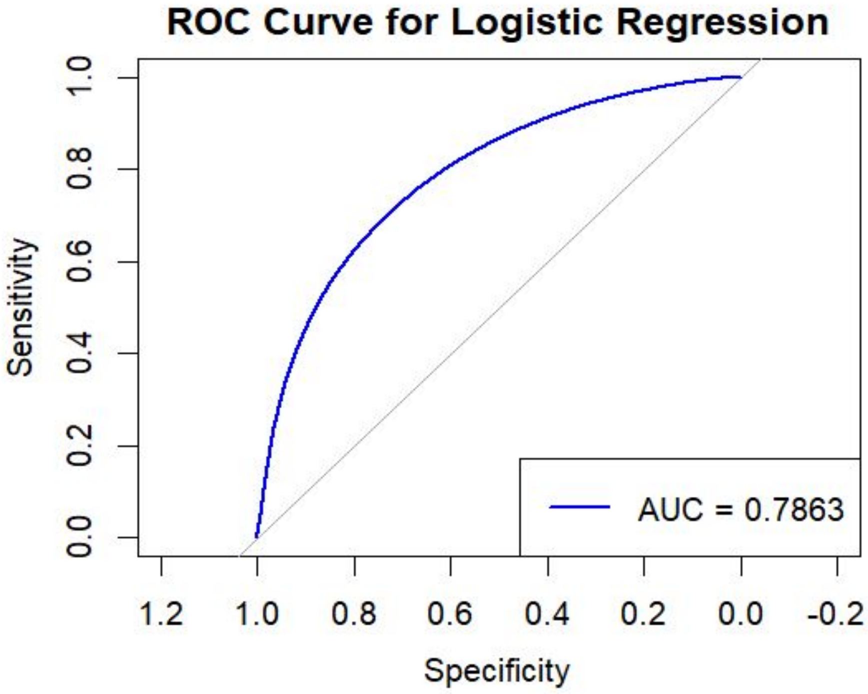
Confusion Matrix

- Sensitivity (**36.32%**) – Model was able to correctly identify non-homeowners
- Specificity (**92.52%**) – Model was able to correctly identify homeowners

	Actual 0	**Actual 1**
Prediction 0	55872	29961
Prediction 1	97957	370551

Data Modeling: Logistic Regression

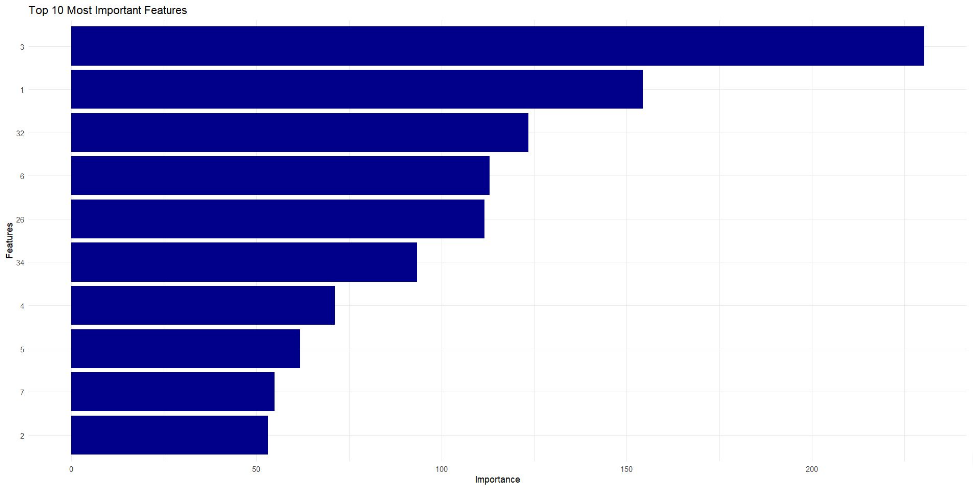
ROC Curve:



Data Modelling: Logistic Regression

Most Influential Features:

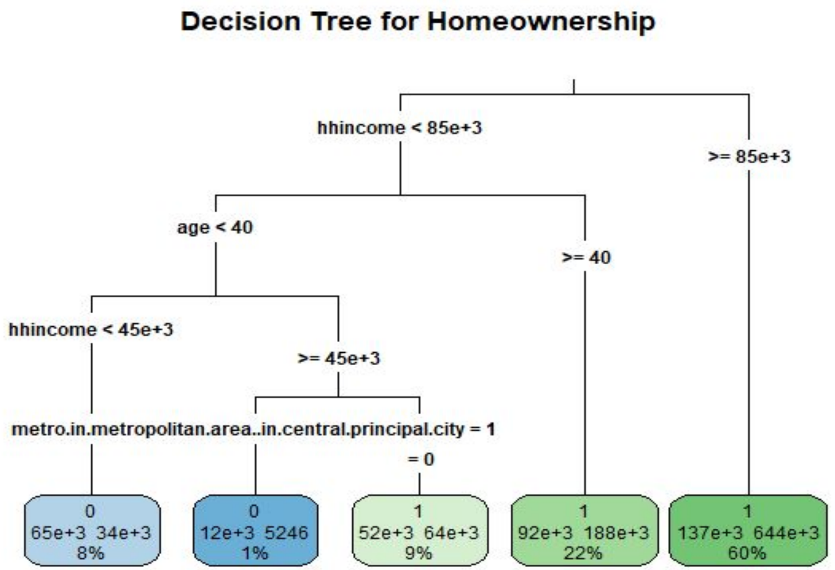
- Household income
- Age
- Citizen Status
- Metropolitan Status
- Education
- Race
- Sex
- Marital Status



Data Modeling: Metrics Comparison

Metric	**Value**	**Interpretation**
Accuracy	76.92%	Overall correctness of the model predictions
AUC	0.7863	The model has good discriminatory power
Sensitivity (Recall)	36.32%	The model struggles to correctly classify 0 (non-homeowners)
Specificity	92.52%	The model is very good at identifying 1 (homeowners)
Precision	65.09%	Among predicted positives, 65% are actual positives
Balanced Accuracy	64.42%	Averages sensitivity and specificity to account for imbalanced data
Kappa	0.3338	Moderate agreement between predictions and actual outcomes
McNemar's Test	p < 2.2e-16	Indicates predictions significantly differ from random guessing

Data Modelling: Decision Tree



Decision Tree AUC: 0.6796

THANKS!

Any questions?

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